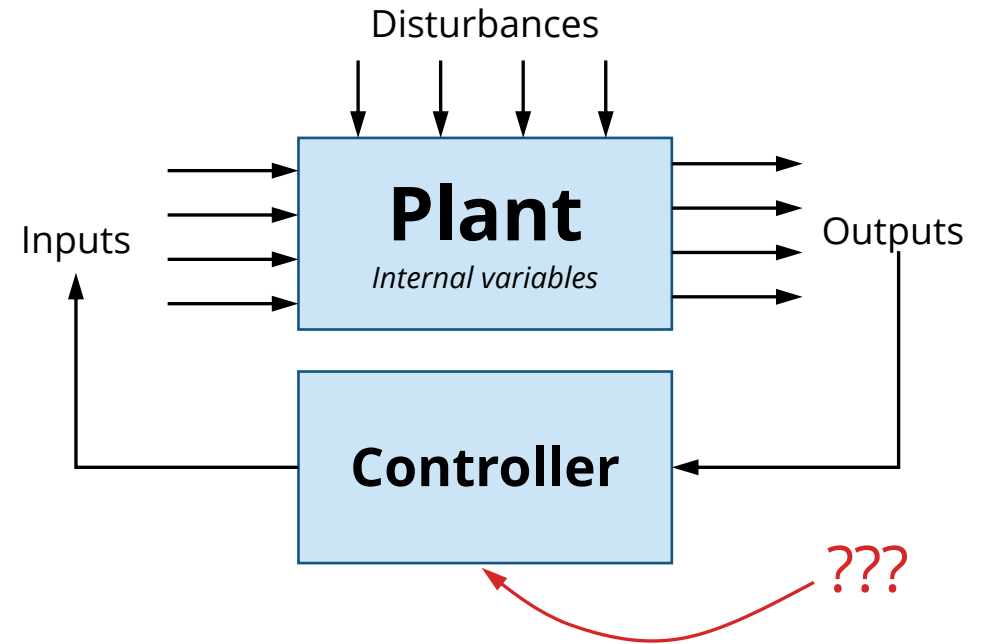




AA 513 Multivariable Control

Lecture 07 // Week 7

Instructor: Prof. Karen Leung

Announcements

- Tutorial topic
 - Better to choose a simpler topic and focus on explaining/visualizing the concepts
 - Don't over-complicate the problem
- Rubrics coming out soon

Last time

- HJB equation – continuous time, continuous state and controls
- Linear quadratic regulator
 - Problem: quadratic cost, linear dynamics, no other constraints
 - **Discrete** time, **finite** horizon: perform Riccati recursion
 - **Discrete** time, **infinite** horizon: Discrete Algebraic Riccati equation (DARE)
 - **Continuous** time, **finite** horizon: solve ODE
 - **Continuous** time, **infinite** horizon: Continuous Algebraic Riccati equation (CARE)
 - Variants: time-varying LQR, iterative LQR, tracking LQR,

Finite vs infinite time

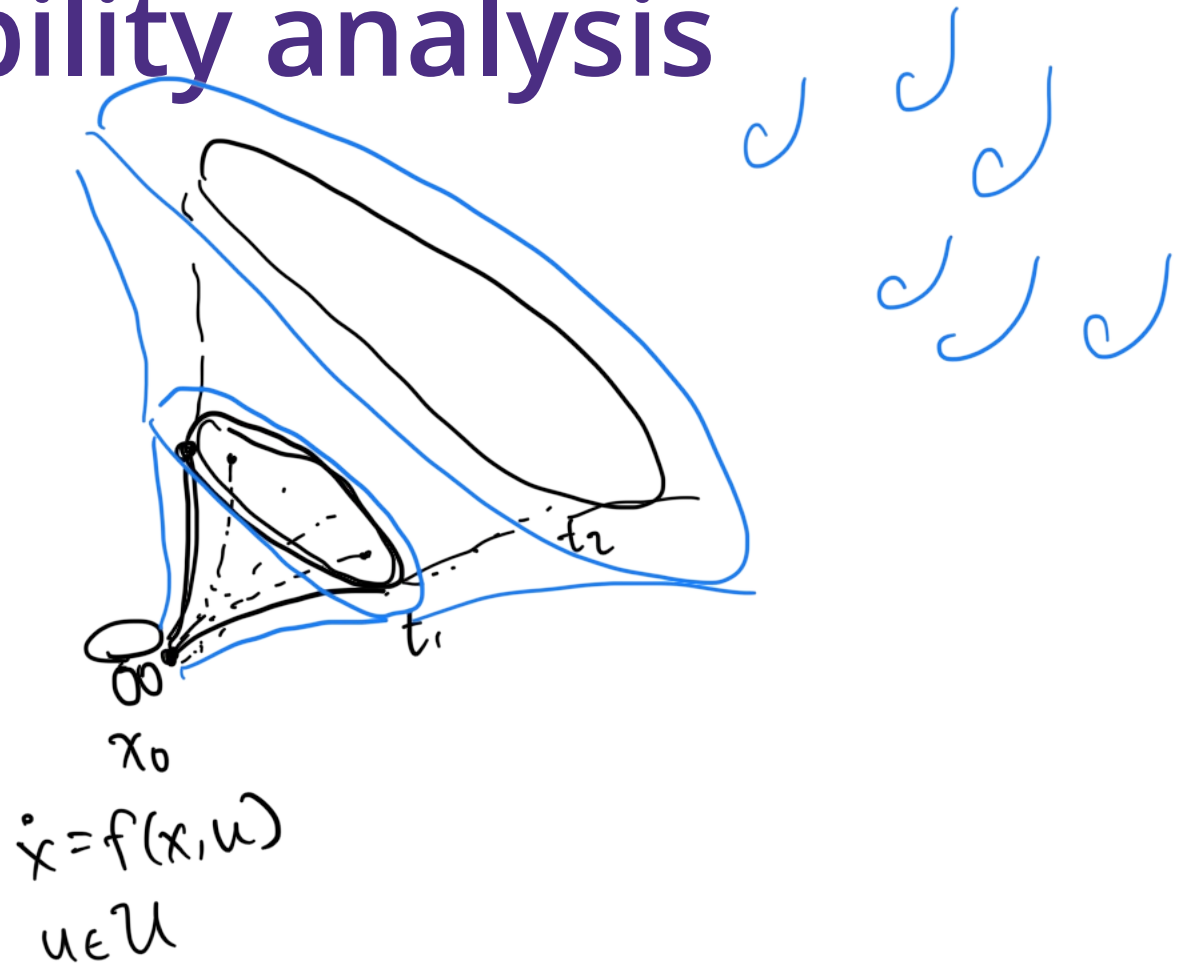
- Infinite time—when there is no “deadline” to reach the goal
 - Time-varying does not apply
 - Optimal gain is **constant**
- Finite time—want to reach the goal within a fixed time
 - Time-varying is possible
 - Optimal gain is **time-varying**
 - If far from goal and little time left, there is trade-off between control effort to get to goal quickly and not reaching goal

This week

- Reachability analysis
- HJ Reachability

Reachability analysis

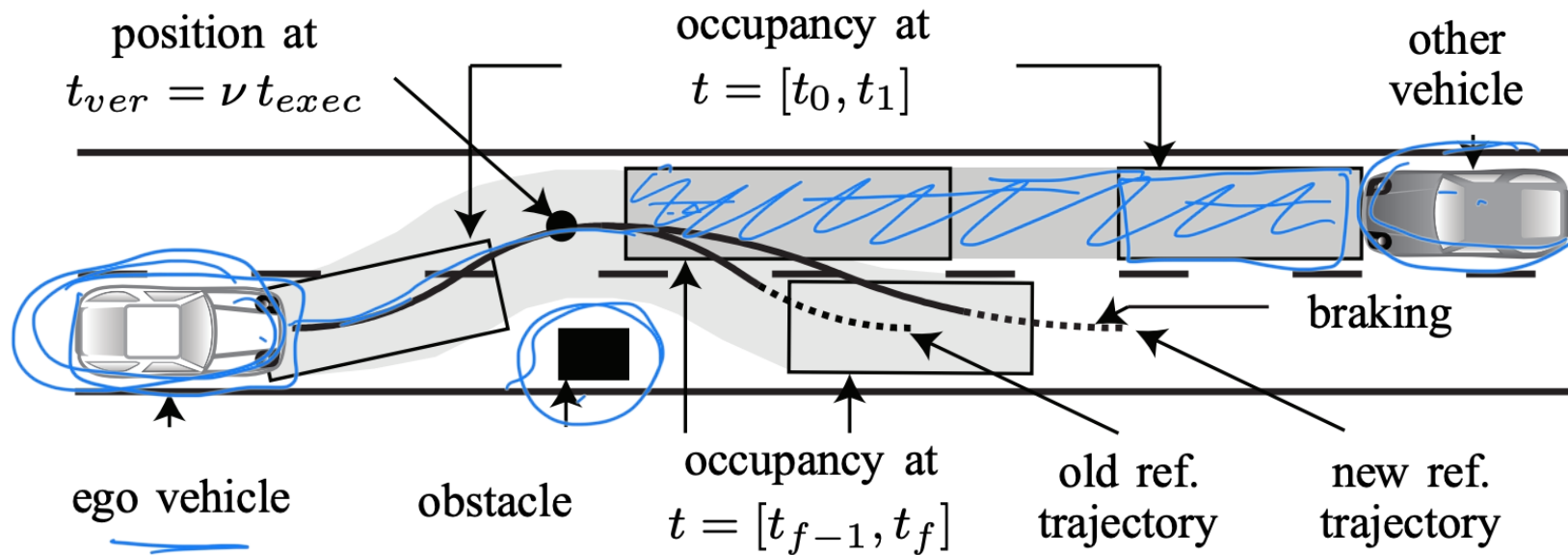
- The study of the set of states a dynamical system can (and can't) reach over time given:
 - dynamics
 - initial conditions,
 - possible controls
 - disturbances



Reachability analysis: Applications

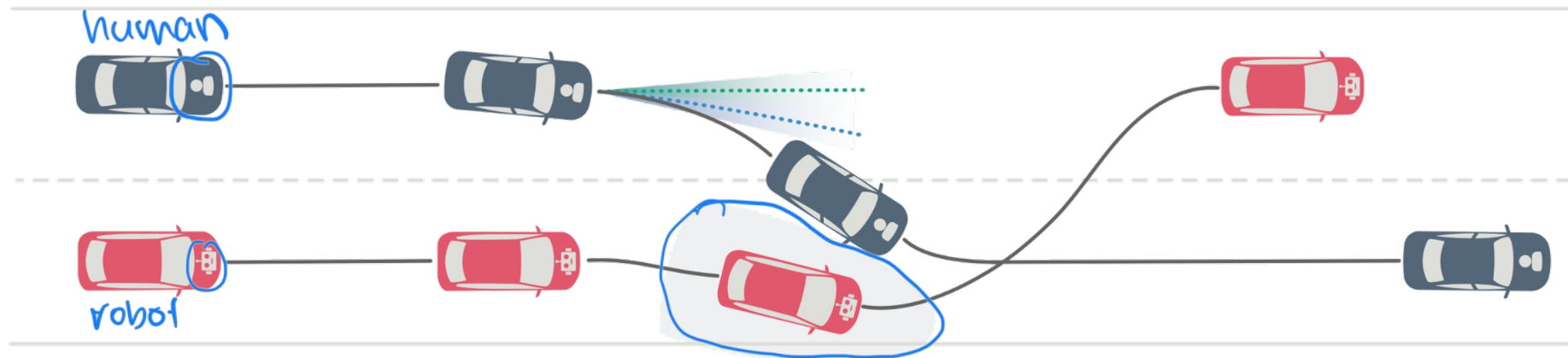
- Safety-critical control: Avoid inevitable collision states
- Robust control: Plan a “tube path” to avoid obstacles
- Provide guarantees: Certify safe/unsafe states before deployment

Reachability analysis: Applications



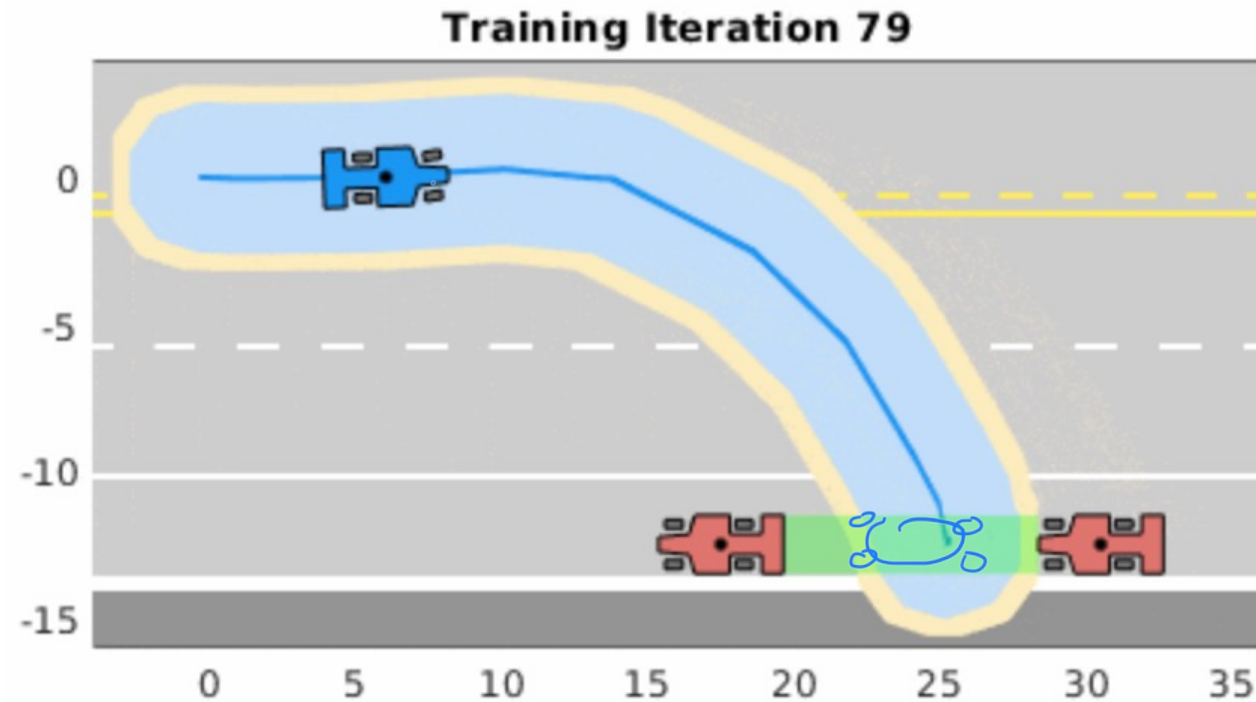
M. Althoff and J. M. Dolan, "Online Verification of Automated Road Vehicles Using Reachability Analysis," in *IEEE Transactions on Robotics*, vol. 30, no. 4, pp. 903-918, Aug. 2014,

Reachability analysis: Applications



Leung, K., Schmerling, E., Zhang, M., Chen, M., Talbot, J., Gerdes, J. C., and Pavone, M., *On Infusing Reachability-Based Safety Assurance within Planning Frameworks for Human-Robot Vehicle Interactions*, *Int. Journal of Robotics Research*, 2020

Reachability analysis: Applications



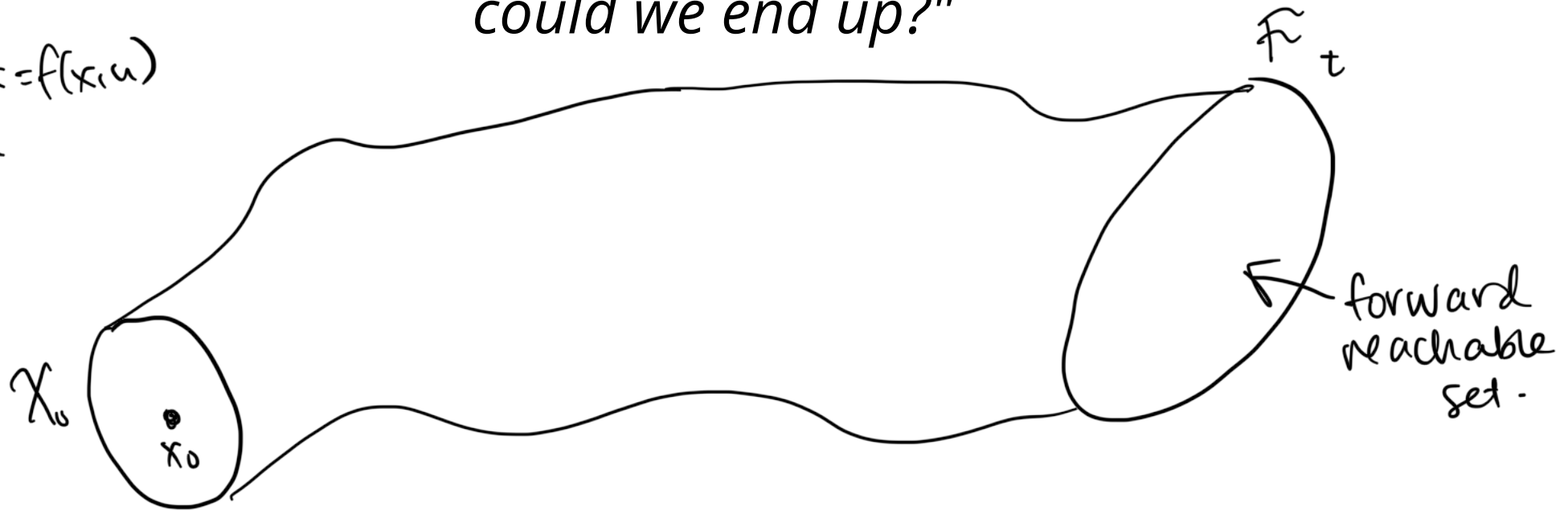
Long Kiu Chung, Shreyas Kousik, *Provably-Safe Neural Network Training Using Hybrid Zonotope Reachability Analysis*, Conference on Decision and Control, 2025

Forward reachability (open-loop)

Forward reachability: The set of states a system can reach within some time using any control input

"Starting from here, under any allowed input/disturbance, where could we end up?"

given $\dot{x} = f(x, u)$
 $u \in \mathcal{U}$



Forward reachability (open-loop)

Forward reachability: The set of states a system can reach within some time using any control input

"Starting from here, under any allowed input/disturbance, where could we end up?"

Open loop because:

- Assume all possible control inputs are applied without correction along the way. I.e., no feedback.

$$\mathcal{F}(x_0, t) = \{x \in \mathcal{X} \mid \exists u(\cdot) \in \mathbb{U}[0, t] \mid (x) = \xi_{x_0, 0}^{u(\cdot)}(t)\}$$

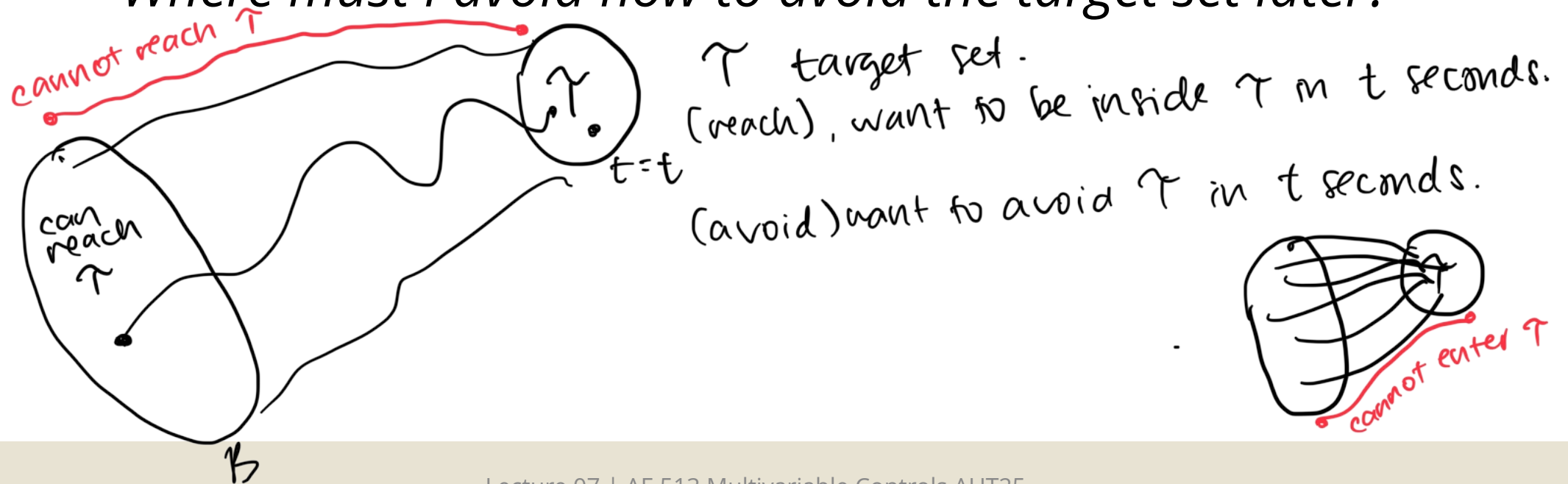
time (pointing to t)
 current state (pointing to x_0)
 all $x \in \mathcal{X}$ s.t. - there exists a control signal (under the $\mathbb{U}[0, t]$ and (x))
 solving the $\dot{x} = f(x, u)$ with ctrl signal $u(\cdot)$ (above the $\xi_{x_0, 0}^{u(\cdot)}(t)$)

Backward reachability (closed-loop)

Backward reachability: Given a *target set*, what are all the states that could reach (or avoid) that set in the future?

"Where must I be now to reach the target set later?"

"Where must I avoid now to avoid the target set later?"



Backward reachability (closed-loop)

Backward reachability: Given a *target* set, what are all the states that could reach (or avoid) that set in the future?

“Where must I be now to reach the target set later?”

“Where must I avoid now to avoid the target set later?”

Closed-loop because

- At each state and time, we consider the best control to take to reach/avoid the target set

Sounds like dynamic programming...

Backward reachable sets

Reach case

target set, want to reach
/ horizon

there exist a control signal

state at time t from executing $u(\cdot)$ w/ dyn. $\dot{x} = f(x, u)$

$$\mathcal{R}(\mathcal{T}, t) = \{ \underline{x_0} \in \mathcal{X} \mid \exists u(\cdot) \in \mathbb{U}[0, t] \text{ s.t. } \underline{\xi_{x_0, 0}^{u(\cdot)}}(t) \in \mathcal{T} \}$$

Avoid case (Inevitable collision set)

target set
want to avoid
/ horizon

end up in $\mathcal{T}$ at time t .

$$\underline{\mathcal{A}}(\mathcal{T}, t) = \{ x_0 \in \mathcal{X} \mid \forall u(\cdot) \in \mathbb{U}[0, t] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(t) \in \mathcal{T} \}$$

How to compute BRS?

- This is an optimal control problem! But we only care about whether we reach/avoid target set at the **end** of the horizon

min
 $u(\cdot) \in \mathcal{U}[0, T]$
 control signal
 subject to

$\Rightarrow g=0$
 running cost

$$\int_0^T g(x, u, t) dt + \boxed{g_T(x(T))}$$

terminal cost.

only care what happens at
 the end of time -
 do we enter $\uparrow$?
 or avoid $\uparrow$?

$$\dot{x} = f(x, u, t) \text{ dynamics}$$

$$x_0 = x(0) \text{ current state}$$

$$u \in \mathcal{U}(x), x \in \mathcal{X}$$

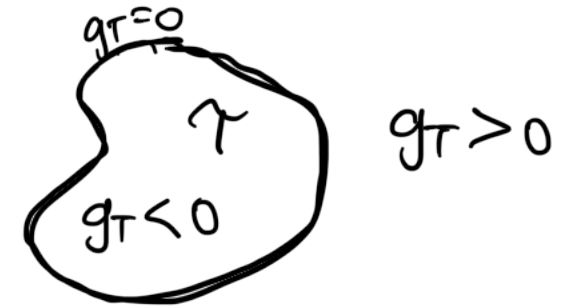
feasible control & state.

Define target set & value function interpretation

Consider reach case. Being inside target set is desirable

$$\mathcal{T} = \{x \in \mathcal{X} \mid \underline{g_T}(x) \leq 0\}$$

eg. signed distance function



Cost-to-go

$$V(x, t) = \min_{u(\cdot) \in \mathcal{U}[t, T]} \underline{g_T}(x(T))$$

value function

Do the optimal thing to get inside target set as far as possible

in the reach case, we want to be inside $\mathcal{T}$. $\Rightarrow$ want g_T small

in the avoid case, we want to be outside $\mathcal{T}$ $\Rightarrow$ want g_T large.

What about the **avoid** case? instead of $\min_u$ we want to $\max_u$.

Going back to HJB equation (BRS)

$$\text{HJB} \quad \frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \{ \cancel{g(\mathbf{x}(t), \mathbf{u}(t), t)} + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \} = 0$$

Terminal condition: $V(\mathbf{x}, T) = g_T(\mathbf{x})$

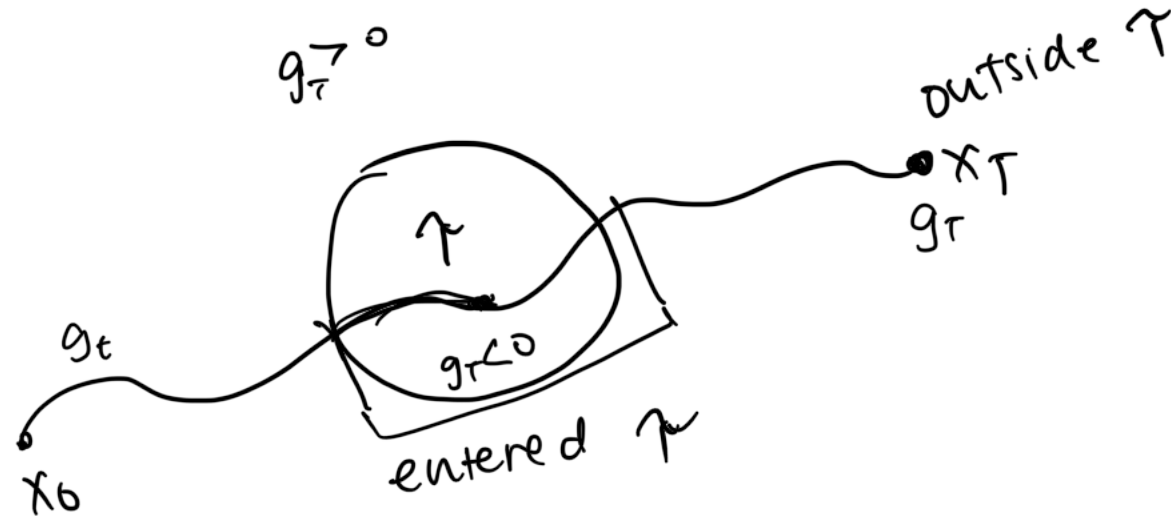
Zero running cost! $g(\mathbf{x}, \mathbf{u}, t) = 0$

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) = 0$$

$$V(\mathbf{x}, T) = g_T(\mathbf{x})$$

Solve this PDE...

Wait a minute...what if trajectories *enter* the target set, but then escape it afterwards?



$$\mathcal{T} = \{x \in \mathcal{X} \mid g_T(x) \leq 0\} \quad V(x, t) = \min_{\underline{u(\cdot)} \in \mathbb{U}[0, T]} \left(\min_{s \in [0, T]} \underline{g_T(x(s))} \right)$$

Lowest cost over horizon

Backward reachable tube

Reach case

$$\tilde{\mathcal{R}}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists \underline{s} \in [0, t], \exists u(\cdot) \in \mathbb{U}[0, \underline{s}] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(\underline{s}) \in \mathcal{T}\}$$

Avoid case (Inevitable collision set)

$$\tilde{\mathcal{A}}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists \underline{s} \in [0, t], \forall u(\cdot) \in \mathbb{U}[0, \underline{s}] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(\underline{s}) \in \mathcal{T}\}$$

Going back to HJB equation (BRT)

BRS:

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) = 0$$

BRT:

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \underbrace{\min(0, \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t))}_{\text{Hamiltonian}} = 0$$

Idea: Find lowest value. So we can “freeze” value ($\frac{\partial V}{\partial t} = 0$) if system is moving away from target set ($\min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \geq 0$)

Summary so far

- Backward reachability useful for *safety analysis*
 - Is it possible to reach/avoid a set within some time?
- We can use the HJB equation to help us solve some backward reachability problems
 - Zero running cost. Value function has a clear interpretation.
- Reach case:

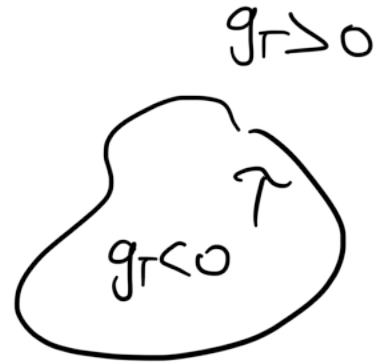
BRS:

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) = 0$$

BRT:

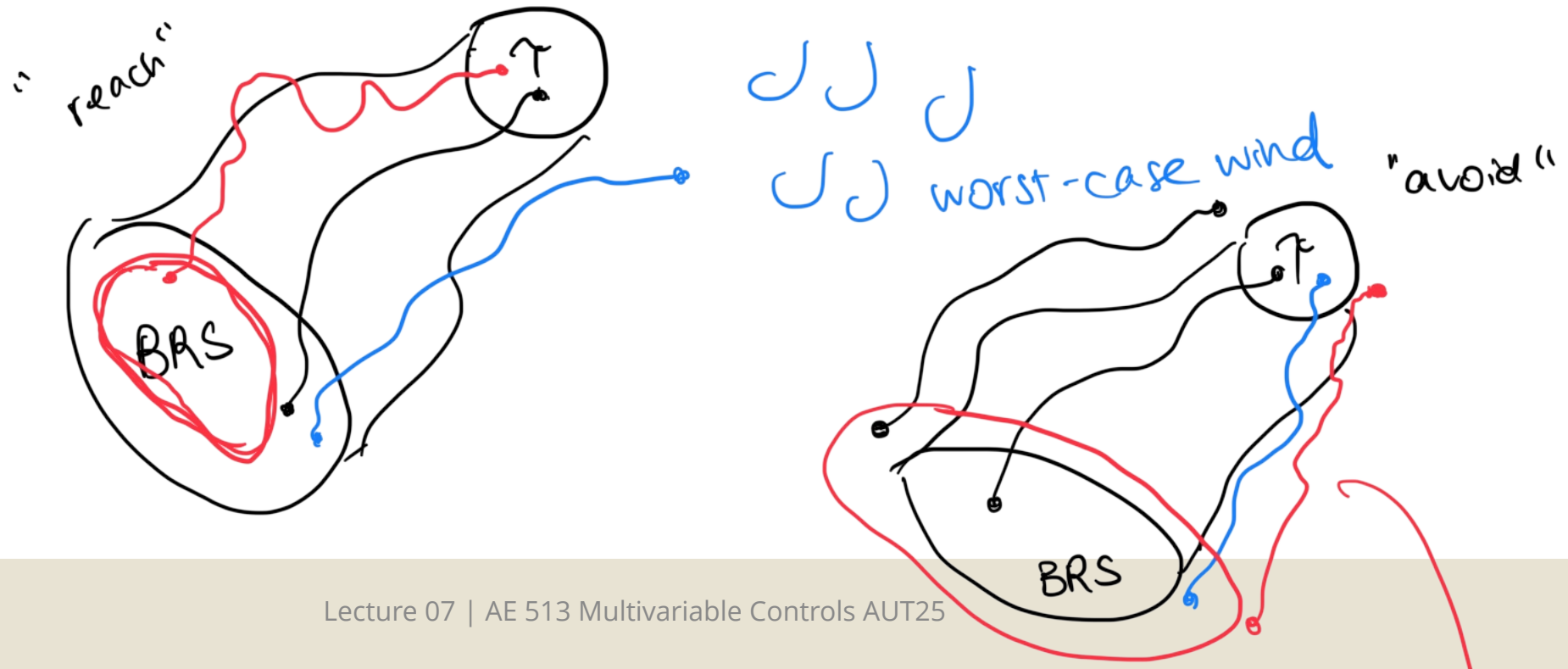
$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min(0, \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)) = 0$$

- Avoid case: $\max_{\mathbf{u}(t) \in U(\mathbf{x}(t))}$



What about disturbances?

- Consider disturbances in the dynamics $\dot{x} = f(x, u, \textcolor{blue}{d}, t)$
- We would like to be **robust** against any and all disturbances
 - Can we still **reach** the target set despite worst-case disturbances?



What about disturbances?

- Consider disturbances in the dynamics $\dot{x} = f(x, u, d, t)$
 - We would like to be **robust** against any and all disturbances
 - Can we still avoid target set despite worst-case disturbances?
- 

What about disturbances?

Perform reachability analysis assuming **adversarial** disturbances

- If control wants to *reach* target, then disturbance aims to *avoid* target
- If control wants to *avoid* target, then disturbance aims to *reach* target

this results in a "game"

player 1 wants to minimize "control" & player 2 wants to maximize "disturbance"

With disturbances

Reach case

$$\Rightarrow \underset{\text{set}}{\mathcal{R}}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \underline{\forall d(\cdot) \in \mathbb{D}[0, t]}, \exists u(\cdot) \in \mathbb{U}[0, t] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(t) \in \mathcal{T}\}$$

$$\underset{\text{tube}}{\widetilde{\mathcal{R}}}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists s \in [0, t], \underline{\forall d(\cdot) \in \mathbb{D}[0, s]}, \exists u(\cdot) \in \mathbb{U}[0, s] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(s) \in \mathcal{T}\}$$

Avoid case

$$\mathcal{A}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists d(\cdot) \in \mathbb{D}[0, t], \forall u(\cdot) \in \mathbb{U}[0, t] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(t) \in \mathcal{T}\}$$

$$\widetilde{\mathcal{A}}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists s \in [0, t], \exists d(\cdot) \in \mathbb{D}[0, s], \forall u(\cdot) \in \mathbb{U}[0, s] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(s) \in \mathcal{T}\}$$

Ordering of control and disturbances

- Consider BRS-reach

(saw this before)

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \underbrace{\min_{\mathbf{u}(t) \in U(\mathbf{x}(t))}}_{\text{only control.}} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) = 0$$

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \max_{\mathbf{d}(t) \in D(\mathbf{x}_t)} \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t) = 0$$

or

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \max_{\mathbf{d}(t) \in D(\mathbf{x}_t)} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t) = 0$$

- Which one should we pick? disturbance "goes second"

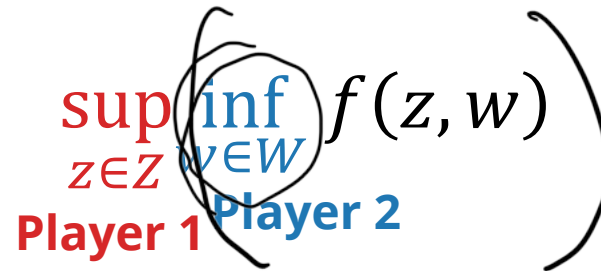
Max-min inequality

For any function $f: Z \times W \rightarrow R$,

$\sup \Rightarrow \max$
 $\inf \Rightarrow \min$

$$\sup_{z \in Z} \inf_{w \in W} f(z, w) \leq \inf_{w \in W} \sup_{z \in Z} f(z, w)$$

- **Interpretation**



The diagram shows the expression $\sup_{z \in Z} \inf_{w \in W} f(z, w)$ with a large right parenthesis. The $\sup$ and $z \in Z$ are in red, and the $\inf$ and $w \in W$ are in blue. A circle is drawn around the $\inf_{w \in W}$ part. Below the red text is the label "Player 1" in red, and below the blue text is the label "Player 2" in blue.

- **Takeaway**

- Acting second has the advantage!

Hamilton-Jacobi-Isaacs Equation captures the "game" element

HJ Reachability with disturbances

- **BRS, reach, with disturbances**

$$\frac{\partial V}{\partial t}(x(t), t) + \min_{u(t) \in U(x(t))} \max_{d(t) \in D(x_t)} \nabla V(x(t), t)^T f(x(t), u(t), d(t), t) = 0$$

- **BRT, reach, with disturbances**

$$\frac{\partial V}{\partial t}(x(t), t) + \min(0, \min_{u(t) \in U(x(t))} \max_{d(t) \in D(x_t)} \nabla V(x(t), t)^T f(x(t), u(t), d(t), t)) = 0$$

- **BRS, avoid, with disturbances**

$$\frac{\partial V}{\partial t}(x(t), t) + \max_{u(t) \in U(x(t))} \min_{d(t) \in D(x_t)} \nabla V(x(t), t)^T f(x(t), u(t), d(t), t) = 0$$

- **BRT, avoid, with disturbances**

$$\frac{\partial V}{\partial t}(x(t), t) + \min(0, \max_{u(t) \in U(x(t))} \min_{d(t) \in D(x_t)} \nabla V(x(t), t)^T f(x(t), u(t), d(t), t)) = 0$$

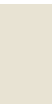
What if we have control-disturbance-affine dynamics?

- Suppose we have dynamics $\dot{\mathbf{x}} = f_0(\mathbf{x}) + B_u(\mathbf{x})\mathbf{u} + B_d(\mathbf{x})\mathbf{d} \Leftarrow$
- Assume control and disturbances are *bounded*
 - $\mathbf{u} \in [\mathbf{u}_{\min}, \mathbf{u}_{\max}]$, $\mathbf{d} \in [\mathbf{d}_{\min}, \mathbf{d}_{\max}]$
- Consider the term $\nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t)$
- Is solving for the optimal control/disturbance tractable?

$$\begin{aligned}
 & \min_{\mathbf{u} \in \mathcal{U}} \max_{\mathbf{d} \in \mathcal{D}} \nabla V(\mathbf{x}, t)^T (f_0(\mathbf{x}) + B_u(\mathbf{x})\mathbf{u} + B_d(\mathbf{x})\mathbf{d}) \\
 &= \min_{\mathbf{u} \in \mathcal{U}} \max_{\mathbf{d} \in \mathcal{D}} \nabla V(\mathbf{x}, t)^T f_0(\mathbf{x}) + \nabla V(\mathbf{x}, t)^T B_u(\mathbf{x})\mathbf{u} + \nabla V(\mathbf{x}, t)^T B_d(\mathbf{x})\mathbf{d} \\
 &= \underbrace{\nabla V(\mathbf{x}, t)^T f_0(\mathbf{x})}_{\text{constant given } \mathbf{x} \text{ \& } t} + \min_{\mathbf{u} \in \mathcal{U}} \underbrace{\nabla V(\mathbf{x}, t)^T B_u(\mathbf{x})\mathbf{u}}_{\text{constant}} + \max_{\mathbf{d} \in \mathcal{D}} \underbrace{\nabla V(\mathbf{x}, t)^T B_d(\mathbf{x})\mathbf{d}}_{\text{constant}}
 \end{aligned}$$

Bang-bang control

$\mathbf{u}^*, \mathbf{d}^*$ will be the extreme values of $\mathcal{U}$ & $\mathcal{D}$



Computing HJ reachable sets

- https://github.com/StanfordASL/hj_reachability
- https://github.com/StanfordASL/hj_reachability/blob/main/examples/quickstart.ipynb
-

```
dynamics = hj.systems.Air3d()

grid = hj.Grid.from_lattice_parameters_and_boundary_conditions(hj.sets.Box(np.array([-6., -10., 0.]), np.array([20., 10., 2 * np.pi])), (51, 40, 50), periodic_dims=2)

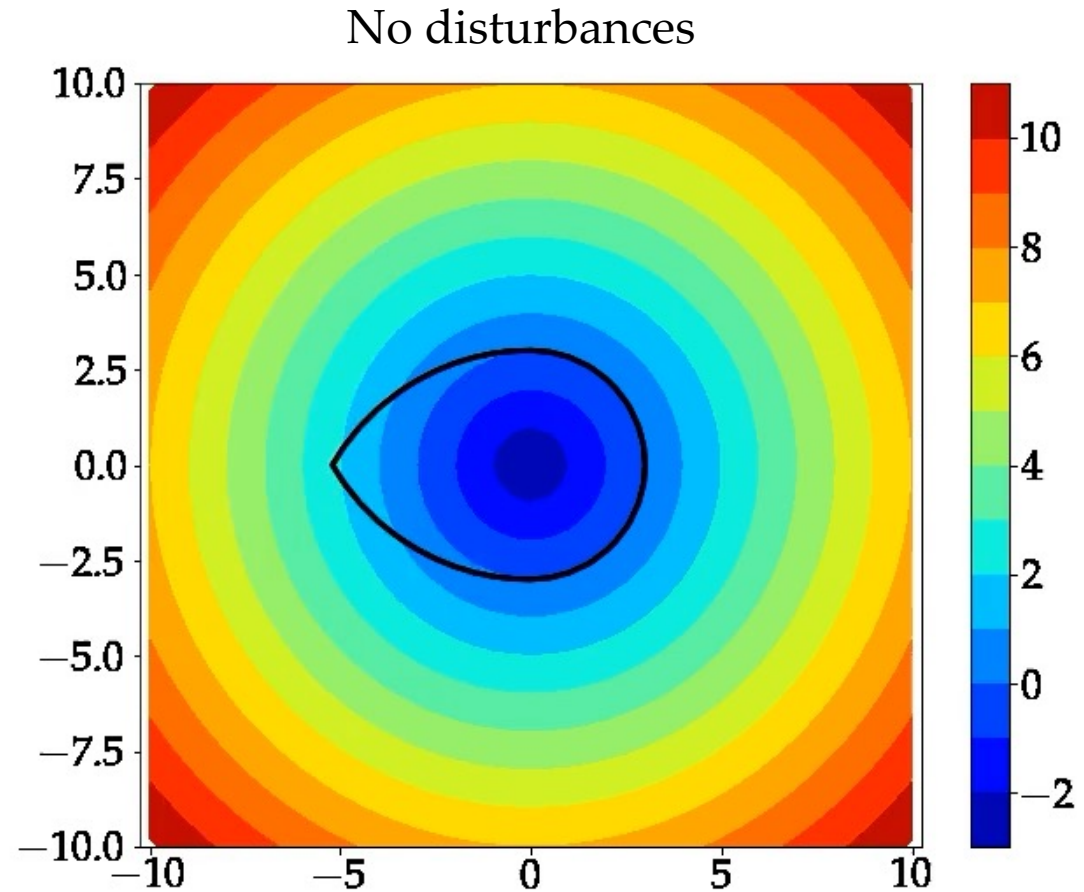
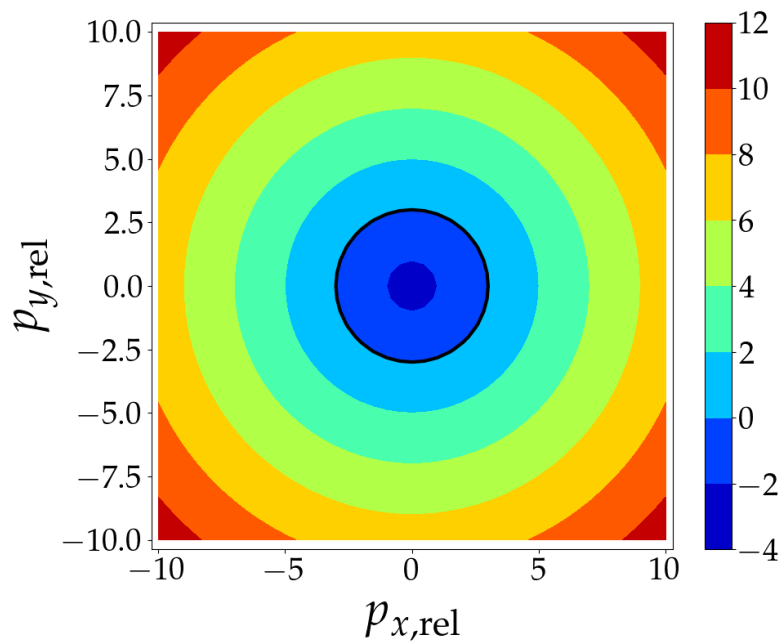
values = jnp.linalg.norm(grid.states[:, :, :2], axis=-1) - 5

solver_settings = hj.SolverSettings.with_accuracy("very_high",
hamiltonian_postprocessor=hj.solver.backwards_reachable_tube)
```

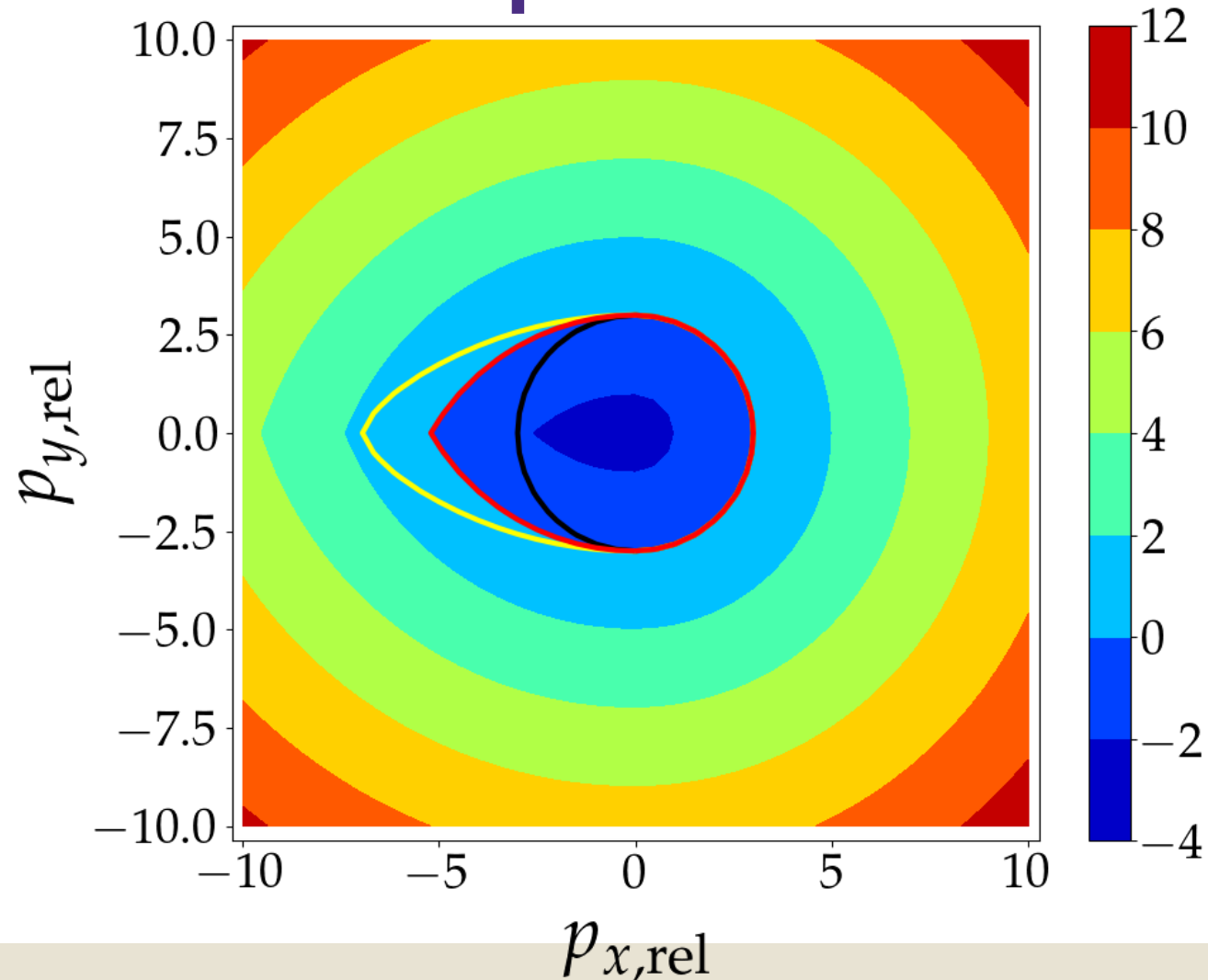
Dubins car example (BRT, avoid)

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v \cos \theta \\ v \sin \theta \\ u + d \end{bmatrix}$$

Constant velocity
Can only control yaw rate



Dubins car example with disturbances

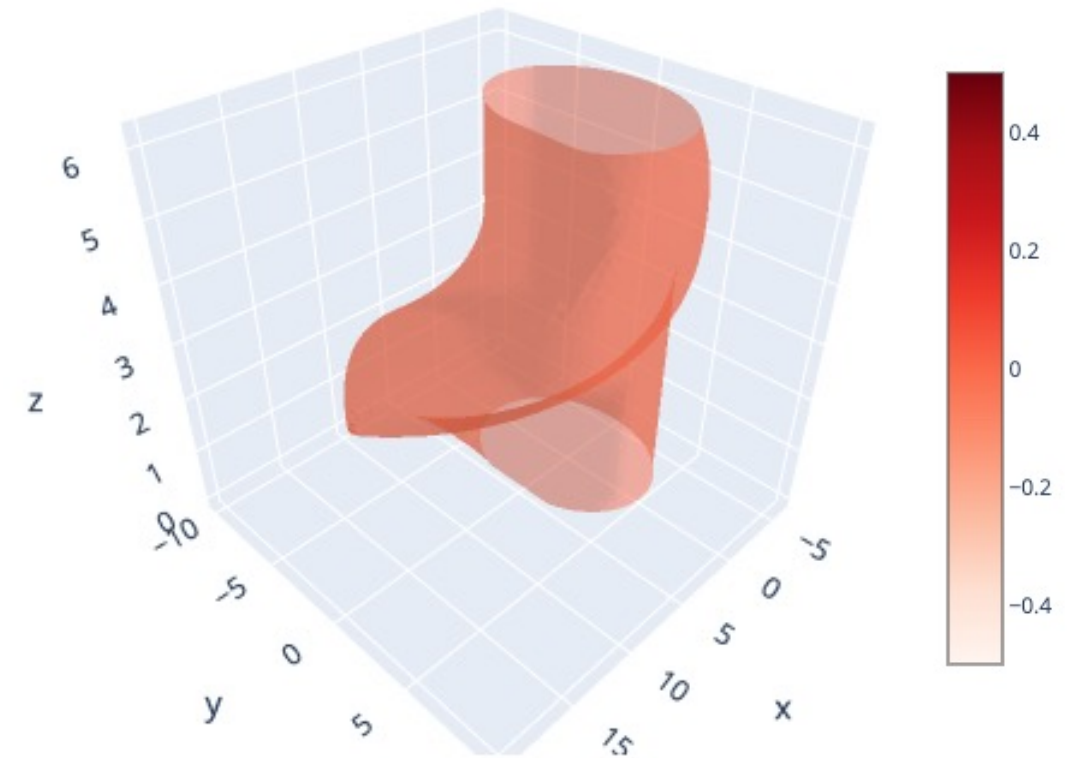
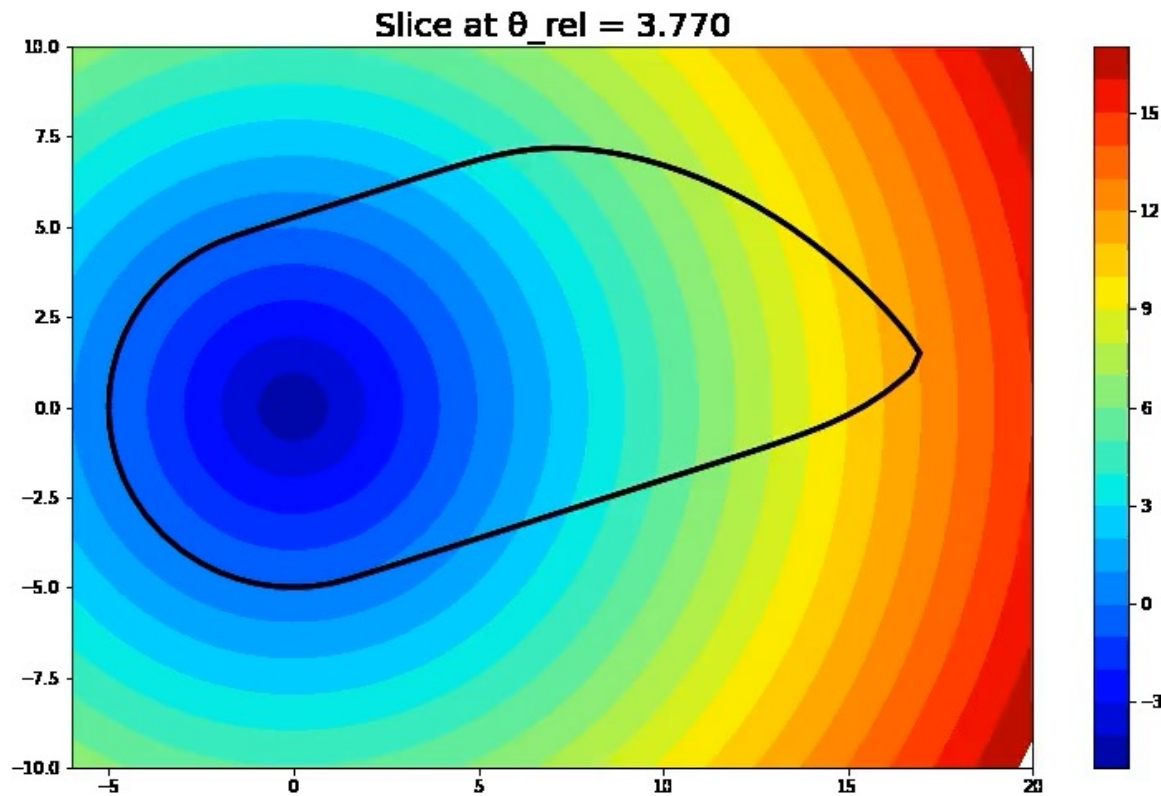


Pursuit-evasion example

- Can consider a *relative* system between two agents & treat other agent as a disturbance
- Consider two Dubins' car agents, one is pursuing the other (i.e., cat and mouse)

$$\begin{bmatrix} \dot{x}_{\text{rel}} \\ \dot{y}_{\text{rel}} \\ \dot{\theta}_{\text{rel}} \end{bmatrix} = \begin{bmatrix} -v_a + v_b \cos \theta_{\text{rel}} + y_{\text{rel}} u_a \\ v_b \sin \theta_{\text{rel}} - x_{\text{rel}} u_a \\ u_b - u_a \end{bmatrix}$$

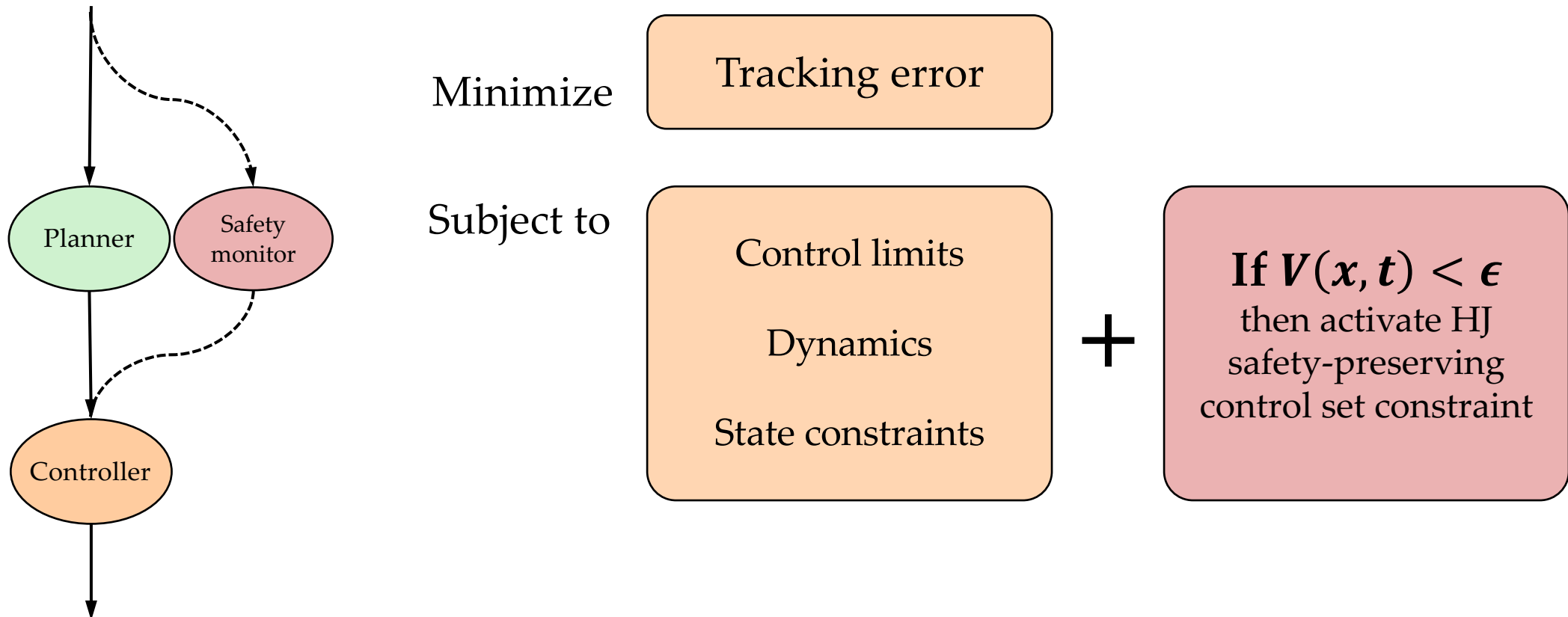
Pursuit-evasion example



What can I do with the HJ value function?

- HJ value functions are CBFs!
 - <https://arxiv.org/abs/2104.02808>
 - <https://youtu.be/9SIO-oNT7Gs>
- Least-restrictive controller
 - When $V(x, t) \leq \epsilon$, switch to optimal HJ policy
- Minimally-interventional safe controller
 - When $V(x, t) \leq \epsilon$, add HJ control constraint in tracking controller
 - <https://arxiv.org/abs/2012.03390>

Minimally-interventional control



Leung, K., Schmerling, E., Zhang, M., Chen, M., Talbot, J., Gerdes, J. C., and Pavone, M., *On Infusing Reachability-Based Safety Assurance within Planning Frameworks for Human-Robot Vehicle Interactions*, *Int. Journal of Robotics Research*, 2020

